

Solution Requirements

Date	4 july 2026
Team ID	xxxxxxx
Project Name	Credit card approval
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users must log in securely using a unique username and password to access the credit card prediction form. describe how users will securely log in or verify their identity</i>	Medium
2	Authorization levels	<i>Regular users can submit application details and view their results. Admin users can access a dashboard to see overall system evaluation metrics.</i>	high
3	External interfaces	<i>The application interface connects directly with the backend machine learning model file (model.pkl) to</i>	High

		<i>calculate prediction scores automatically.</i>	
4	Transactions processing	<i>The system must securely process user-submitted data parameters (like income and credit history records) and send them to the Flask backend without data loss.</i>	High
5	Reporting	<i>The application must display an instantaneous dashboard status page (result.html) with clear data charts representing approval or rejection metrics.</i>	Medium
6	Business rules, etc.	<i>The application must follow strict checking logic: any application with negative background history parameters or incomplete data records is flagged immediately.</i>	High
7	Compliance to laws or regulations	<i>Regular users can submit application details and view their results. Admin users can access a dashboard to see overall system evaluation metrics.</i>	High
8	Other	<i>The application interface connects directly with the backend machine learning model file (model.pkl) to calculate prediction scores automatically.</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The web application interface must process the user inputs and return the machine learning approval prediction response in less than 2 seconds.</i>	<i>(e.g., Page loads in < 2 seconds)</i>
2	Scalability	<i>The Python Flask backend environment must be built to support multiple concurrent loan applicants submitting forms at the same time without crashing.</i>	<i>(e.g., Supports up to 10,000 concurrent users)</i>
3	Security & Data Privacy	<i>All input data records (like salaries or identification metrics) must be safely handled during execution to prevent external data leaks.</i>	<i>(e.g., End-to-end AES-256 encryption)</i>
4	Reliability & Availability	<i>The prediction model server must maintain high uptime so users can check their card approval eligibility at any time of day.</i>	<i>(e.g., 99.9% uptime per month)</i>
5	Usability & Accessibility	<i>The user interface form fields must use clear text labels and straightforward navigation so beginners can easily fill out their parameters.</i>	<i>(e.g., Complies with WCAG 2.1 AA standards)</i>
6	Other	<i>The system code structure should be documented cleanly so that it remains easily maintainable and portable to cloud servers later.</i>	